

African Economic Stress & Early-Warning System

A data-driven framework for measuring economic stress, testing early-warning signals, forecasting growth and stress-testing Kenya against external shocks

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Economic stress in these eight African economies is measurable and trackable, but it cannot be forecast from annual indicators: the stress index follows real episodes, while the early-warning model scored an AUC of 0.55, barely above a coin flip.

1. Executive summary

The project delivers a working stress-monitoring framework and one honest negative result: the index describes past stress well, but no tested model predicts next-year growth slumps. All figures use the corrected data described in Section 3.

Question	Evidence	Verdict
Can indicators be combined into a stress index?	Kenya peaks at 51.6 in 2008, 45.1 in 2011 and 42.9 in 2023; region-wide stress peaks in 2008-09 and 2022-24	Yes, for price and currency stress
Does the index predict slumps?	Out-of-sample AUC 0.55 (0.50 = chance); Brier score 0.175 versus 0.172 for always guessing the average	No demonstrated skill
Which model forecasts Kenya's growth best?	Damped-trend ETS, RMSE 2.29 points versus 2.55 for a plain average; difference not significant ($p = 0.43$)	ETS, weakly
What does the forecast say?	5.1% growth in 2026 and 5.1% in 2027, typical one-year error about 1.7 points	A statistical baseline, not an outlook
How exposed is Kenya to shocks?	A 15% depreciation lifts the index from 21.4 to 32.2 (Low to Moderate); a 2-point regional slowdown cuts growth by 1.4 points	Currency and regional growth matter most
How does Kenya compare?	Average index 32.1 versus 51.6 for Ghana and 47.2 for Nigeria;	Lower stress, slower growth

Question	Evidence	Verdict
	growth 4.4% versus 8.5% for Ethiopia	

What to use. The stress index as a descriptive dashboard of inflation and currency pressure; the country profiles and peer comparison; the forecast comparison as a benchmark for any future model.

What not to use. The slump probabilities for decisions. They are not better calibrated than the average slump rate (18%), and the highest-stress country (Nigeria) gets the lowest slump probability.

Why the result is still valuable. The pipeline is reproducible from public data, validation is out of sample, and the failure of the early-warning model is documented with robustness checks rather than hidden. That is what a credible economic analyst is expected to show.

2. Background, research questions and scope

Economic stress is the build-up of pressures, such as rising prices, a weakening currency, external deficits and falling output, that makes a downturn or a crisis more likely. Central banks, finance ministries and investors track such pressures with composite indicators because no single variable captures them.

Why Kenya and these peers. Kenya is a net oil importer, so it is exposed to global fuel prices and to the shilling. In this dataset Kenya's real GDP per person rose from about US\$1,195 in 2000 to about US\$1,908 in 2025 (constant prices), while exports fell from 21.6% to 15.8% of GDP over the same period. The peer set mixes East African neighbours (Uganda, Tanzania, Rwanda, Ethiopia) with larger African economies (Nigeria, Ghana, South Africa) that have different structures, including oil exporters. That mix allows a pooled model with enough observations and a fair comparison.

Types of stress the project considers

Type	Indicator used	What it signals
Price stress	Inflation	Erosion of purchasing power, tighter monetary policy
External and currency stress	Depreciation against the US dollar, current account	Import costs, debt-service pressure, capital flight
Real-economy stress	GDP growth, unemployment	Weak activity and jobs
Fiscal stress	Government debt	Not usable: only 12% coverage (Section 3)

Research questions

1. Can several indicators be combined into one Economic Stress Index that tracks known episodes of stress?
2. Does the index, or its components, predict a growth slump in the following year?
3. Which model forecasts Kenya's GDP growth best, and how accurate is it?

4. How sensitive is Kenya's stress level to oil, inflation, exchange-rate and regional-growth shocks?
5. How does Kenya compare with seven African peers, and why might it differ?

Hypotheses stated before testing

- H1: the index rises in years of high inflation and currency depreciation. Supported (Section 5).
- H2: higher stress today raises the chance of a growth slump next year. Not supported (Section 6).
- H3: models using inflation and oil prices forecast Kenya's growth better than univariate models. Not supported (Section 7).
- H4: an oil-price rise raises Kenya's stress through inflation and depreciation. Not supported in this sample (Section 8).

Scope and exclusions. Annual data, 2000-2025, eight countries. No quarterly or monthly data, no fiscal variables, no policy-rate or reserves data, no causal identification. The oil-price series is global, not Kenya-specific, and the pump-price or fuel-import channel is not modelled.

3. Data

The analysis panel has 208 country-years (8 countries x 26 years, 2000-2025) with no duplicate keys, built from World Bank open data retrieved through the public API.

Sources. World Development Indicators (annual series, retrieved for 1990-2025 and analysed from 2000), and the World Bank Pink Sheet annual nominal crude-oil price (US dollars per barrel).

Variable dictionary

Variable	WDI code	Unit	Role
GDP growth	NY.GDP.MKTP.KD.ZG	% per year	Index input, forecast target
Inflation, consumer prices	FP.CPI.TOTL.ZG	% per year	Index input
Unemployment (ILO modelled)	SL.UEM.TOTL.ZS	% of labour force	Index input
Official exchange rate	PA.NUS.FCRF	LCU per US\$	Source of depreciation (% change, + = weaker currency)
Current account balance	BN.CAB.XOKA.GD.ZS	% of GDP	Index input
Government debt (central)	GC.DOD.TOTL.GD.ZS	% of GDP	Excluded: 12% coverage
Exports, imports	NE.EXP.GNFS.ZS, NE.IMP.GNFS.ZS	% of GDP	Collected, not modelled

Variable	WDI code	Unit	Role
Gross capital formation	NE.GDI.TOTL.ZS	% of GDP	Collected, not modelled
GDP per capita	NY.GDP.PCAP.KD	constant US\$	Collected, not modelled
Crude oil, average	Pink Sheet	US\$ per barrel	Growth rate used as predictor and scenario driver

Coverage in the 2000-2025 window

Variable	Pooled coverage	Notes
GDP growth, inflation, unemployment, oil	100%	Complete
Exchange rate	99%	Missing for Ethiopia and Tanzania in 2025
Current account	93%	Rwanda 58%; others 96-100%
Government debt	12%	Zero for Kenya, Ghana, Nigeria, Rwanda, Tanzania; 27% Ethiopia and Uganda; 42% South Africa

Treatment. Gaps of up to two years inside a series were interpolated; remaining gaps were filled with the country median and flagged in the dataset (current account 7.2% of values). Extreme values were winsorised at the 1st and 99th percentiles before standardising, so that Ghana's 2000 depreciation (104%) and Nigeria's 2024 depreciation (129%) do not dominate the principal components. Debt was dropped automatically because coverage fell below the 60% rule.

A data correction made during this review. The delivered run computed depreciation with a method that carried the last exchange rate forward, so the missing 2025 rates for Ethiopia and Tanzania appeared as 0% depreciation. In this report those two values are instead filled with each country's median depreciation (5.5% and 3.6%) and flagged. The effect is confined to those two countries' 2025 readings (Ethiopia 36.9 to 41.4, Tanzania 25.8 to 28.8); every headline conclusion is unchanged. The pipeline script has been fixed so a fresh run behaves the same way.

Known data cautions

- Unemployment is an ILO modelled estimate. Kenya's series is almost flat (2.6-2.9%) until 2016 and then steps to about 5%, which reflects the model, not necessarily the labour market.
- Exchange-rate series are period averages of official rates, so parallel-market premia (important in some years for Ethiopia, Nigeria and Ghana) are not captured.
- 2025 is the latest year and may be revised by the World Bank.
- Countries differ in structure (oil exporters versus importers, very different unemployment levels), which matters for any pooled model.

4. Methods

The method has five stages, each fixed before looking at results except where Section 3 and Section 6 say otherwise: build the stress index, define a slump target, validate an early-warning classifier, compare forecasting models for Kenya, and run shock scenarios.

4.1 Economic Stress Index

1. Winsorise each indicator at its 1st and 99th pooled percentile.
2. Standardise to z-scores using pooled means and standard deviations.
3. Multiply by a sign so that higher always means more stress: growth and the current account are negated, inflation, unemployment and depreciation are not.
4. Extract the first principal component (PC1) with PCA, oriented so its loadings sum positive.
5. Rescale the PC1 score to 0-100 by the pooled minimum and maximum.

$$z_{ij} = s_j \cdot \frac{(x_{ij} - \mu_j)}{\sigma_j}, ESI_i = 100 \cdot \frac{(PC1_i - \min(PC1))}{(\max(PC1) - \min(PC1))}$$

Bands: Low below 25, Moderate 25-50, High 50-75, Severe 75 and above. The stored means, loadings and limits are reused to re-score scenarios, so a shocked Kenya is measured on the same ruler.

4.2 Slump target and early-warning model

In year t , the target equals 1 if growth in year $t+1$ is below 2% or falls by more than 3 percentage points from year t . Pooled, 36 of 200 labelled observations (18%) are slumps. Two logistic regressions with an L2 penalty ($C = 0.5$) on standardised inputs: model A uses the index alone, model B the five components plus oil-price growth.

$$P(\text{slump}_{t+1} = 1) = 1 - (1 + e^{-(\beta_0 + X_t \beta)})$$

4.3 Validation

- **Expanding window.** For each test year T from 2008 to 2024, train on all labelled rows from years up to $T-1$ and predict year T for all eight countries; pool the 136 predictions.
- **Leave-one-country-out.** Train on seven countries, test on the eighth.
- **Metrics.** AUC (probability a random slump is ranked riskier than a random non-slump; 0.5 is chance) and the Brier score (mean squared probability error; lower is better), compared with the Brier score of always predicting the average slump rate.

4.4 Forecasting Kenya's GDP growth

One-step-ahead forecasts from rolling origins (training ends 2004-2024, giving 21 forecasts for 2005-2025).

Model	Specification
Naive	Last observed growth

Model	Specification
Mean	Expanding historical average
ARIMA	Order chosen by AIC over p 0-2, d 0-1, q 0-2
ETS	Additive damped trend, parameters estimated
VAR(1)	Growth, inflation, oil-price growth
Ridge	Penalised regression (alpha 5) on two growth lags, lagged inflation, lagged oil growth

Accuracy: MAE, RMSE, MAPE (only where actual growth exceeds 0.5% in absolute value, because percentage errors explode near zero) and bias. Models were compared with a Diebold-Mariano-style test on squared errors, without a small-sample correction.

4.5 Scenario engine

Country-fixed-effects regressions on the 200-observation panel: a reduced form (growth on oil growth and the leave-one-out peer average growth), a structural form adding inflation and depreciation, and pass-through regressions of inflation and depreciation on oil growth. Each shock modifies Kenya's 2025 inputs, the index and slump probability are recomputed, and the growth effect comes from the regression coefficients.

4.6 Tools and reproducibility

Python with pandas, NumPy, scikit-learn (PCA, logistic regression, Ridge, metrics), statsmodels (ARIMA, ETS, VAR) and matplotlib, run in Google Colab. A single script downloads the data, caches it, and writes the panel, an Excel workbook, charts and a results file. The separate steps-and-tools report documents each step and contains the code.

5. Results: the Economic Stress Index

The index is an inflation and currency-pressure gauge: it correlates 0.86 with inflation and 0.75 with depreciation across the 208 country-years, but only 0.10 with GDP growth. That makes it good at flagging price and exchange-rate episodes and poor at flagging output collapses.

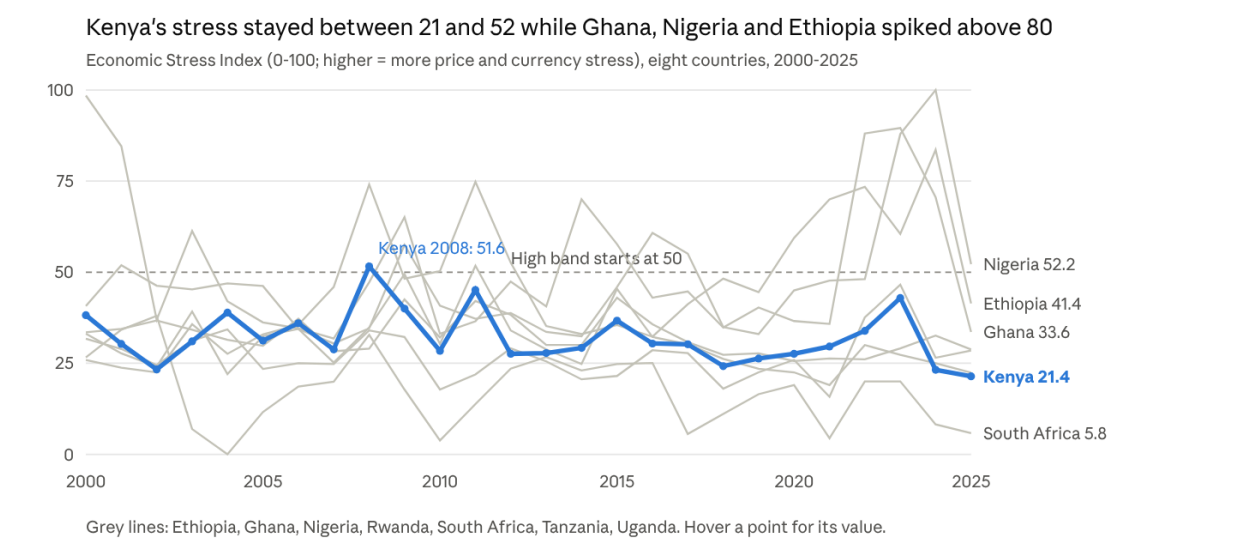
Construction. PC1 captures about 32% of the variance of the five indicators, so stress here has several independent dimensions rather than one dominant factor.

Indicator	PC1 loading	Correlation with index
Inflation	+0.69	0.86
Depreciation	+0.62	0.75
Current account	-0.12	0.15
GDP growth	-0.08	0.10

Indicator	PC1 loading	Correlation with index
Unemployment (level)	-0.35	-0.44

The negative sign on unemployment is a design flaw, not an economic finding. Countries with structurally high unemployment, above all South Africa (32.4% in 2025), are scored as less stressed. Section 6 tests a variant that uses unemployment change instead.

Distribution of the 208 country-years by band. Low 44, Moderate 137, High 20, Severe 7. All seven Severe readings are currency or inflation crises: Ghana in 2000, 2001, 2022 and 2023, Nigeria in 2023 and 2024, and Ethiopia in 2024.



World Bank WDI and Pink Sheet via the project pipeline · 8 countries, 2000-2025 · corrected panel, primary specification

Kenya crossed the High-band line of 50 only once, in 2008. Ghana, Nigeria and Ethiopia each reached Severe levels in at least one year, mostly through currency and inflation spikes.

5.1 Kenya through time

Year	Index	Band	GDP growth %	Inflation %	Depreciation %
2000	38.2	Moderate	0.6	10.0	8.3
2008	51.6	High	0.2	26.2	2.8
2011	45.1	Moderate	5.1	14.0	12.1
2015	36.7	Moderate	5.0	6.6	11.7
2020	27.6	Moderate	-0.3	5.4	4.4
2023	42.9	Moderate	5.7	7.7	18.6

Year	Index	Band	GDP growth %	Inflation %	Depreciation %
2024	23.2	Low	4.7	4.5	-3.6
2025	21.4	Low	4.6	4.1	-4.1

Kenya's index peaks in 2008 (inflation 26.2%), rises again in 2011 (inflation 14.0%, depreciation 12.1%) and in 2023 (depreciation 18.6%), then falls to among its lowest levels of the period in 2024-2025 (21.4 in 2025 is the minimum) as the shilling strengthened and inflation eased to about 4%. The 2020 contraction (-0.3%) barely registers: it was an output shock with moderate prices and currency movements.

Kenya's index was above the average of its seven peers in only 5 of 26 years, and on average 4.1 points below it.

5.2 Stress by period

Country	2000-07	2008-12	2013-19	2020-25
Ghana	53.2	45.9	47.3	59.1
Nigeria	42.8	41.8	42.2	63.5
Ethiopia	31.2	60.0	38.6	64.7
Tanzania	33.2	36.8	32.1	28.1
Kenya	32.2	38.5	29.3	29.8
Uganda	29.8	40.1	30.1	24.4
Rwanda	28.0	27.0	23.5	30.2
South Africa	19.5	18.3	19.0	12.9

Three patterns stand out. First, the 2008-12 window (global food and fuel price shock and its aftermath) raised stress in Ethiopia, Uganda and Kenya. Second, the 2020-25 window split the region: stress rose sharply in Ethiopia, Nigeria and Ghana, where inflation and depreciation surged, but fell in Tanzania, Uganda and South Africa. Third, Kenya sits in the stable middle group with Tanzania, Uganda and Rwanda.

The all-country average index peaked at 44.1 in 2009 and at 50.5 in 2023, with 44.7 in 2022 and 46.2 in 2024; its lowest values were 28.1 in 2018 and 29.3 in 2025.

5.3 Peak stress by country

Country	Peak index	Year	What drove it (from the data)
Nigeria	100.0	2024	Depreciation 129%, inflation 33%
Ghana	98.5	2000	Depreciation 104%
Ethiopia	83.6	2024	Depreciation 51%
Kenya	51.6	2008	Inflation 26%
Uganda	51.8	2011	Inflation 17%, depreciation 16%
Rwanda	46.6	2023	Inflation 20%
Tanzania	43.0	2015	Depreciation 21%
South Africa	38.0	2002	Depreciation 22%, inflation 9.5%

These peaks are consistent with widely reported episodes (Nigeria's exchange-rate reform and Ethiopia's currency liberalisation around 2024), but the data here do not identify policy causes; they show only that price and currency pressure drove the scores.

6. Results: the early-warning model

Neither specification shows useful out-of-sample skill: AUC is 0.55 at best, and both Brier scores are worse than simply predicting the 18% average slump rate for every country-year.

Specification	Test observations	AUC	Brier score	No-skill Brier
A: stress index only	136	0.548	0.1771	0.1719
B: components plus oil growth	136	0.554	0.1754	0.1719

6.1 Country-by-country robustness

In leave-one-country-out tests, AUC ranged from 0.36 (Ghana) to 0.71 (Nigeria), scattering around 0.5 rather than clustering above it.

Held-out country	AUC	Held-out country	AUC
Nigeria	0.71	Rwanda	0.56
South Africa	0.66	Uganda	0.50

Held-out country	AUC	Held-out country	AUC
Kenya	0.64	Ethiopia	0.46
Tanzania	0.46	Ghana	0.36

6.2 What the coefficients say

Odds ratios per one standard deviation, specification B. No standard errors were computed, so none of these should be called significant.

Predictor	Odds ratio	Reading
Unemployment (level)	1.86	Mostly separates South Africa from the rest
GDP growth	1.14	Counter-intuitive sign (weak)
Depreciation	1.07	Negligible
Current account	0.95	Negligible
Inflation	0.81	Counter-intuitive sign
Oil-price growth	0.63	Higher oil growth goes with a lower next-year slump risk, plausibly a global-demand effect

6.3 Why the model fails

- **Few events.** Only 36 slump years among 200 labelled observations, and 11 of them (31%) are South Africa's.
- **The label measures low growth as much as a slump.** A growth threshold of 2% flags a chronically slow economy almost every year. South Africa averaged 2.2% growth and is flagged in 11 of 25 years, against 2 of 25 for Tanzania and Uganda.
- **Annual frequency.** Stress that builds and bursts within a year is invisible, and global shocks (2008-09, 2020) hit all countries at once, so cross-country variation does not help.
- **Stress is mostly price and currency pressure,** while the slump label is about output. The two are only weakly linked (correlation of the index with growth: 0.10).
- **Heterogeneity.** Oil exporters and importers react differently to the same signals.

6.4 Robustness: unemployment change instead of level

Because the unemployment level penalises South Africa mechanically, the index was rebuilt using the yearly change in unemployment. This variant was run on the same panel after seeing the first results, so it is a robustness check, not a new primary specification.

Variant	AUC (A)	AUC (B)	Brier (B)	South Africa index	Kenya index
Unemployment level (primary)	0.548	0.554	0.1754	5.8	21.4
Unemployment change	0.407	0.458	0.1925	17.9	11.4

Fixing the South Africa artefact does not rescue prediction: AUC falls below 0.5. The conclusion that these indicators do not predict next-year slumps does not depend on this choice.

6.5 Latest readings (2025 data, risk for 2026)

Country	Stress index	Band	Slump probability	Index if unemployment change is used
Nigeria	52.2	High	10%	46.4
Ethiopia	41.4	Moderate	17%	24.0
Ghana	33.6	Moderate	12%	25.3
Tanzania	28.8	Moderate	17%	15.6
Rwanda	28.5	Moderate	34%	14.1
Uganda	22.4	Low	18%	7.8
Kenya	21.4	Low	19%	11.4
South Africa	5.8	Low	65%	17.9

The index ranking (Nigeria highest) and the slump probabilities (South Africa highest) point in opposite directions, which is itself evidence that the two outputs measure different things. Ethiopia's and Tanzania's 2025 values use a filled exchange-rate figure (Section 3), so they are the least certain. Use the index for monitoring and do not act on the probabilities.

7. Results: forecasting Kenya's GDP growth

Damped-trend ETS had the lowest RMSE (2.29 points) over 21 rolling one-step forecasts, but no model beats it with statistical confidence, so the ranking is indicative only. Kenya's growth averaged 4.4% over 2000-2025 (standard deviation 2.1, range -0.3% to 8.1%).

Model	MAE	RMSE	MAPE %	Bias	Within 1 point	Test p-value vs ETS
ETS (damped trend)	1.66	2.29	23.1	+0.17	48%	n/a
Expanding mean	2.14	2.55	34.4	-1.50	29%	0.43
VAR(1) growth-inflation-oil	1.93	2.74	26.5	-1.07	48%	0.28
ARIMA (AIC grid)	1.80	2.77	22.7	-0.44	57%	0.18
Ridge (ML benchmark)	2.18	2.90	32.9	-1.29	33%	0.19
Naive (last value)	2.05	3.03	29.6	+0.02	52%	0.17

7.1 Reading the comparison

- **ARIMA is best in ordinary years, ETS in shock years.** ARIMA's median absolute error is only 0.49 points, but a few large misses push its RMSE to 2.77. ETS's smoother forecasts lose less when growth jumps.
- **Adding inflation and oil did not help.** The VAR, which uses them, ranks third and below the plain average, and the Ridge model that also uses oil ranks fifth. This is consistent with the weak oil results in Section 8.
- **Complexity did not win.** The machine-learning benchmark ranks below the plain historical average on RMSE, which is typical for short annual series.
- **Bias.** The mean, VAR and Ridge forecasts are biased low (average error between -1.1 and -1.5 points) (the expanding mean in particular carries the weak early-2000s years); ETS and the naive forecast are nearly unbiased.
- **Statistical caution.** With 21 test points, a p-value of 0.17 to 0.43 against ETS means the differences could easily be noise.

7.2 Where every model failed

Year	Actual growth %	ETS forecast %	Error (points)
2020	-0.3	5.4	+5.6
2008	0.2	5.3	+5.0
2010	8.1	4.4	-3.7

Year	Actual growth %	ETS forecast %	Error (points)
2021	7.6	4.6	-3.0
2005	5.9	3.0	-2.9

The largest misses are the 2008 and 2020 shocks and the rebounds that followed (2010 and 2021). No annual statistical model anticipates those; they need scenario analysis and judgement, not curve fitting.

7.3 The 2026-2027 projection

ETS projects real GDP growth of **5.1% in 2026 and 5.1% in 2027** (5.10% and 5.13%). The forecast is close to a flat line at Kenya's recent average, because the damped trend leaves no momentum to extrapolate. As a yardstick for uncertainty, the model's typical one-year error is about 1.7 points (MAE) and its RMSE is 2.3 points, so outcomes between roughly 3% and 7% would be unremarkable. This is an informal range, not a statistical confidence interval, and shocks like 2008 or 2020 would fall far outside it.

Treat the figure as a statistical baseline against which official or institutional forecasts can be compared, not as a prediction of the macroeconomic outlook.

8. Results: scenario analysis for Kenya

A 15% shilling depreciation is the only shock that moves Kenya out of the Low stress band, from 21.4 to 32.2, while the regional-growth link is the only relationship in the data strong enough to trust. Baseline inputs are Kenya's 2025 values: growth 4.6%, inflation 4.1%, unemployment 5.4%, currency appreciation of 4.1%, current account -4.2% of GDP and oil-price growth of -14.4%.

Scenario	Growth change (points)	Stress index	Change	Band	Slump probability	Change (points)
Baseline	0.00	21.4	0.0	Low	19.5%	0.0
Oil price +30%	+0.19	20.1	-1.3	Low	12.8%	-6.7
Inflation +3 points	-0.12	24.8	+3.4	Low	18.2%	-1.3
Shilling depreciates 15%	-0.19	32.2	+10.7	Moderate	20.7%	+1.2
Regional growth -2 points	-1.40	20.9	-0.5	Low	18.6%	-0.9

Under the robustness variant (unemployment change in the index) the ordering is the same: depreciation raises the index from 11.4 to 23.7, inflation to 15.0, oil lowers it to 9.6 and a regional slowdown raises it to 12.7, and all stay in the Low band.

8.1 Regression evidence behind the scenarios

Country-fixed-effects regressions on 208 country-years. t-statistics use classical standard errors, so true uncertainty is larger than shown.

Relationship	Coefficient	t-statistic	Reading
Regional (peer) growth to growth	0.70	6.5	Strong: 1 point of regional growth goes with 0.7 points at home
Oil-price growth to growth	0.0065	1.0	Not distinguishable from zero
Inflation to growth	-0.042	-1.4	Negative, not significant
Depreciation to growth	-0.013	-0.7	Not distinguishable from zero
Oil-price growth to inflation	0.014	0.9	No detectable pass-through
Oil-price growth to depreciation	-0.085	-3.3	Higher oil goes with less depreciation

8.2 Interpretation

- **Regional growth is the dependable channel.** A 2-point regional slowdown goes with about a 1.4-point slowdown in Kenya, yet the index barely moves because it gives growth little weight. This is the clearest example of the index missing real-economy stress.
- **No evidence that oil hurts here.** Oil-price growth is positively linked to growth and negatively to depreciation. Oil prices tend to rise in global expansions, when African currencies are firmer, and the panel includes oil exporters (Nigeria, Ghana). The data cannot isolate the importer channel that matters for Kenya.
- **The currency scenario is largely mechanical.** Depreciation is one of the index's two main inputs, so a 15% shock mechanically adds 10.7 points. It does not say how the currency would move.
- **The inflation scenario has an odd probability effect.** The slump probability falls slightly because the fitted inflation coefficient is negative, an artefact of the weak model, not a finding.
- **Scenario effects on slump probability should not be used.** They inherit the early-warning model's lack of skill.

8.3 What the scenarios can and cannot say

They can show how the index responds to defined shocks and which inputs dominate it. They cannot predict how Kenya's economy would respond: they are associations estimated on annual data from eight countries, each shock is applied alone, and no policy response, fuel-price pass-through, reserves or fiscal channel is modelled.

9. Country profiles and peer comparison

Kenya combines low-to-moderate stress with slower growth than its East African neighbours: it ranks second-lowest of eight on 2025 stress and second-lowest on average growth. The comparison below covers 2000-2025 unless stated.

Country	Avg growth %	Growth SD	Worst year	Avg inflation %	Peak inflation %	Avg depreciation %	Avg current account % GDP	Avg index	2025 index
Ethiopia	8.5	3.4	-2.2 (2003)	14.8	44.4 (2008)	10.2	-5.8	46.5	41.4
Rwanda	7.5	3.0	-2.8 (2020)	7.0	19.8 (2023)	5.9	-10.6	27.1	28.5
Tanzania	5.9	1.3	2.0 (2020)	6.2	16.0 (2012)	5.2	-5.3	32.4	28.8
Uganda	5.9	2.1	3.0 (2020)	5.9	16.6 (2011)	3.8	-5.4	30.6	22.4
Ghana	5.6	2.7	0.5 (2020)	17.6	41.5 (2001)	17.7	-5.0	51.6	33.6
Nigeria	4.8	4.0	-6.4 (2020)	14.3	33.2 (2024)	13.5	+4.5	47.2	52.2
Kenya	4.4	2.1	-0.3 (2020)	8.3	26.2 (2008)	2.5	-4.2	32.1	21.4
South Africa	2.2	2.5	-6.2 (2020)	5.2	9.9 (2008)	5.0	-2.0	17.6	5.8

The 2025 cross-section

Country	Growth %	Inflation %	Unemployment %	Depreciation %	Current account % GDP
Ethiopia	9.8	13.2	3.3	5.5 (filled)	-5.5
Rwanda	9.4	5.9	11.4	8.4	-10.7
Uganda	6.3	3.6	2.7	-4.1	-6.0
Ghana	6.0	14.2	3.0	-11.5	+8.2
Tanzania	5.9	3.3	1.6	3.6 (filled)	-4.2

Country	Growth %	Inflation %	Unemployment %	Depreciation %	Current account % GDP
Kenya	4.6	4.1	5.4	-4.1	-4.2
Nigeria	4.0	23.0	3.1	2.7	+4.8
South Africa	1.1	3.2	32.4	-2.4	-0.4

Negative depreciation means the currency strengthened against the US dollar.

9.1 Profiles

Kenya. Growth averaged 4.4% with a standard deviation of 2.1, the second-lowest volatility, tied with Uganda and behind Tanzania. It has the lowest average depreciation of the eight (2.5%), but a persistent current account deficit (4.2% of GDP on average) and a 2008 inflation spike of 26.2%. Its index exceeded the peer average in only 5 of 26 years. Three of 25 labelled years are slumps (the slumps themselves were 2002, 2008 and 2020). In 2025 growth was 4.6% with inflation at 4.1%.

Ethiopia. The fastest grower (8.5% average, 9.8% in 2025) and also one of the most stressed: inflation averaged 14.8% and peaked at 44.4% in 2008, and the 51% depreciation of 2024 pushed the index to 83.6. Its average index rose from 31.2 in 2000-07 to 64.7 in 2020-25. Two slump years.

Ghana. The highest average stress (51.6), driven by inflation averaging 17.6% and depreciation averaging 17.7% a year, with Severe readings in 2000, 2001, 2022 and 2023. By 2025 the cedi had strengthened 11.5%, the current account was in surplus (8.2% of GDP) and the index had fallen to 33.6. Growth never turned negative in the sample (minimum 0.5% in 2020).

Nigeria. The most stressed economy in 2025 (index 52.2, inflation 23.0%) and the only one at 100 in the sample (2024, depreciation 129%). Average growth fell from 7.7% in 2000-07 to 1.7% in 2020-25. It is the only country with an average current account surplus (+4.5% of GDP) and had seven slump years.

Rwanda. Fast growth (7.5% average, 9.4% in 2025) with low-to-moderate stress (27.1), but the widest current account deficit (10.6% of GDP on average, 10.7% in 2025) and unemployment of 11.4% in 2025. Inflation reached 19.8% in 2023, giving its peak index of 46.6.

South Africa. The slowest grower (2.2% average; 0.5% in 2020-25) with the lowest index (17.6), but the index is flattered by the unemployment-level artefact: unemployment is 32.4%. It has by far the most slump-labelled years (11 of 25) and the worst single year (-6.2% in 2020).

Tanzania. The steadiest economy: growth never fell below 2.0% (standard deviation 1.3) and the index stayed in a narrow band, peaking at 43.0 in 2015 when the shilling lost 20.5% of its value. Inflation was 3.3% in 2025. Two slump years. The 2025 depreciation figure is filled.

Uganda. Growth averaged 5.9% with a minimum of 3.0% (2020) and low inflation (5.9%), with a stress peak of 51.8 in 2011 (inflation 16.6%, depreciation 15.9%). Stress has since declined (40.1 in 2008-12 to 24.4 in 2020-25). Two slump years.

9.2 Why Kenya differs from its peers

- **Lower exposure to extreme episodes.** Kenya never recorded the triple-digit depreciations seen in Ghana and Nigeria or the 40%-plus inflation seen in Ethiopia and Ghana, so its average index sits 14 to 20 points below theirs.
- **Slower growth than East African neighbours.** Kenya's 4.4% trails Ethiopia (8.5%), Rwanda (7.5%) and Tanzania and Uganda (5.9%), a gap of 1.5 to 4.1 points a year. Only South Africa grew slower.
- **A thinner trade and investment base.** In this dataset Kenya's investment share of GDP was 17.0% in 2025, down from 22.1% in 2015, and its export share fell from 21.6% in 2000 to 15.8% in 2025. These facts are consistent with slower growth but do not prove it.
- **A stable middle position.** Together with Tanzania, Uganda and Rwanda, Kenya forms a group with moderate, low-volatility stress; the volatile group is Ghana, Nigeria and Ethiopia.

10. Conclusions, limitations and next steps

The evidence supports using the index to monitor price and currency stress and does not support using these indicators to predict slumps; the next gains come from better data and a better target, not a more complex model.

10.1 Answers to the research questions

Question	Answer
1. Can an index track stress?	Yes for price and currency stress: it flags Kenya in 2008, 2011 and 2023 and the regional peaks of 2008-09 and 2022-24. It misses output shocks such as 2020.
2. Does it predict next-year slumps?	No. Out-of-sample AUC 0.55 at best, Brier score worse than the no-skill baseline, and a variant scored lower.
3. Best Kenya forecaster?	Damped-trend ETS (RMSE 2.29), not significantly better than a plain average. Projection about 5.1% for 2026 and 2027.
4. Sensitivity to shocks?	The index responds most to depreciation (+10.7 points for 15%). The only reliable growth link is to regional growth (0.70). No adverse oil effect is detectable.
5. Versus peers?	Second-lowest 2025 stress (21.4) and second-lowest average growth (4.4%); far less extreme than Ghana, Nigeria and Ethiopia.

10.2 Limitations

- **Short annual panel.** 26 years and 8 countries give 36 slump events and wide uncertainty on every metric; global shocks (2008-09, 2020, 2022-23) overlap across countries.
- **Target definition.** A fixed 2% growth threshold flags chronically slow economies, so the label partly measures low trend growth.
- **Index design.** It under-weights growth, penalises high-unemployment economies in the primary version, and lacks fiscal variables because debt data were too sparse.
- **Data quality.** Unemployment is modelled, 7.2% of current-account values and two 2025 exchange-rate values are filled, and parallel-market exchange rates are not captured.
- **Statistical inference.** No clustered standard errors, no standard errors on the logistic coefficients, and an informal forecast-comparison test with 21 observations.
- **Pooling.** One set of coefficients for all countries ignores differences between oil exporters and importers.
- **Scenarios are associations.** Shocks are applied one at a time with no policy response or fuel-price channel.
- **Choices made after seeing results.** The unemployment-change variant and the exchange-rate correction came after the first run; both are disclosed and neither changes a conclusion.

10.3 Recommendations

1. **Use the index as a monitoring dashboard** for inflation and currency stress, with Kenya shown against the peer average. In this sample, High or Severe readings (50 and above) cover 27 of 208 country-years (13%), which is a sensible first alert level to review.
2. **Do not publish or act on the slump probabilities** until a model beats the no-skill baseline out of sample.
3. **Improve the index.** Use unemployment change or drop it, add government debt from a fuller source (for example IMF series), add reserves and a lending or policy rate, and give growth more weight so output shocks register.
4. **Improve the target.** Test currency-crisis and inflation-spike definitions, which this index is better suited to anticipate than growth slumps, and growth below each country's own trend.
5. **Raise the data frequency.** Monthly or quarterly exchange-rate, inflation, oil and activity series would multiply the observations and make genuine early warning possible.
6. **Separate oil exporters from importers**, cluster standard errors by country, and add a fuel-price channel for Kenya.

10.4 Roadmap

Step	What	Payoff	Effort
1	Re-run with the corrected script and unemployment change; publish the CSV and Excel outputs	Reproducible baseline	Low

Step	What	Payoff	Effort
2	Publish the interactive dashboard (Overview, Stress Monitor, Forecast, Scenarios, Peer Comparison) as a public web page	Presentable deliverable	Medium
3	Add debt, reserves and a rates series from IMF and central-bank sources	Fiscal and external stress covered	Medium
4	Redefine the target (currency or inflation crisis) and re-validate	Possibly real predictive skill	Medium
5	Move to monthly Kenya data with a nowcasting model	Timely monitoring	High

10.5 Key messages for a policy audience

- Kenya's stress is low in 2025 (21.4) after peaks in 2008, 2011 and 2023, which came from inflation and the shilling.
- The shilling is the most important stress channel in the index; a 15% depreciation would move Kenya into the Moderate band.
- Regional growth is the most reliable external link for Kenya's own growth, so developments in neighbouring economies matter.
- A statistical baseline puts growth near 5.1% in 2026 and 2027, with typical one-year errors of about 1.7 points and no ability to anticipate shocks.
- Kenya's relative problem is growth, not stress: it grows more slowly than East African peers while keeping prices and the currency steadier.

These are analytical findings from public data, not policy advice.

10.6 How to describe this project

"I built a reproducible early-warning pipeline from World Bank data for eight African economies. A PCA-based stress index tracks Kenya's 2008, 2011 and 2023 price and currency episodes. I tested whether it predicts next-year growth slumps using expanding-window validation and found it does not, then checked the result against an alternative specification. I compared six forecasting models for Kenya's growth, found damped-trend smoothing best but not significantly so, and stress-tested Kenya against oil, inflation, currency and regional shocks. The main lesson was that careful out-of-sample validation can show an appealing indicator has no predictive value."

10.7 Glossary

Term	Meaning
AUC	Probability the model ranks a random slump above a random non-slump; 0.5 is chance, 1.0 is perfect
Brier score	Mean squared error of predicted probabilities; lower is better
PCA	Method that finds the weighted combination of indicators capturing the most shared variation
ARIMA, ETS, VAR	Time-series models: autoregressive and moving-average, exponential smoothing, and vector autoregression
RMSE, MAE, MAPE	Forecast error measures: root mean squared, mean absolute, mean absolute percentage
Expanding window	Validation that trains only on the past at each test date
Fixed effects	Regression that removes each country's average level before estimating relationships
Winsorising	Capping extreme values at set percentiles
Depreciation	Percentage rise in local currency per US dollar; negative means the currency strengthened

10.8 Appendix: specification, corrections and reproducibility

- **Primary specification.** Index inputs: growth, inflation, unemployment (level), depreciation, current account. Slump rule: next-year growth below 2.0% or a fall of more than 3.0 points. Logistic regression, $C = 0.5$. Test years 2008-2024. Forecast origins 2004-2024. Shocks: oil +30%, inflation +3 points, depreciation +15%, regional growth -2 points.
- **Corrections applied in this version.** (1) Missing 2025 exchange rates for Ethiopia and Tanzania were treated as 0% depreciation in the delivered run and are now filled with country medians and flagged; this changed Ethiopia's 2025 index from 36.9 to 41.4 and Tanzania's from 25.8 to 28.8, and shifted other 2025 values by 0.1 to 0.5 points through the PCA refit. (2) Unemployment-change robustness variant added. Forecast results are unaffected because they use only Kenya's series.
- **Data vintage.** Series as retrieved from the World Bank API in October 2026; 2025 values may be revised.
- **Code.** The pipeline script and the separate steps-and-tools report contain the full code, the commands to run it and a step-by-step method. The script now uses the corrected exchange-

rate calculation and, by default, the unemployment-change specification; to reproduce this report's primary figures exactly, use unemployment in levels as noted in the steps report.